Lesson 24: Taylor Series

Assessment Criteria Rubric

An analytical rubric for Taylor Series frequently breaks down assessment into these areas:

- **Derivation and Formula Application (25-35%):** Correct calculation of derivatives ($f'(a), f''(a), \ldots, f^{(n)}(a)$) and proper substitution into the Taylor formula $\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$.
- **Expansion and Notation (15-20%):** Accuracy in writing out terms, handling factorial terms, and using sigma notation correctly.
- **Radius and Interval of Convergence (20-25%):** Application of the Ratio Test or Root Test to determine the range of x values for which the series converges.
- **Error Approximation and Remainder (15-20%):** Using Taylor's Inequality (Lagrange Error Bound) or the Alternating Series Estimation Theorem to determine the maximum error of the approximation.
- **Conceptual Application (10-15%):** Using known series (e.g., e^x, $\sin x$) to find new series, differentiating/integrating term-by-term, or approximating non-polynomial functions.